

Union and Intersection of Sets

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The **union of two sets**, written as $A \cup B$, is the set containing all elements that belong to A , B , or both.

Union of Sets

The **union of two sets** contains all elements that belong to either set or both.

$$A = \{1, 2, 3\}$$

$$B = \{3, 4, 5\}$$

Therefore,

$$A \cup B = \{1, 2, 3, 4, 5\}$$

Intersection of Sets

The **intersection of two sets**, written as $A \cap B$, is the set containing only the elements common to both sets.

$$A = \{1, 2, 3\}$$

$$B = \{3, 4, 5\}$$

Therefore,

$$A \cap B = \{3\}$$

Elements in the Union

Elements that appear in both sets are written only once in the union.

$$A = \{1, 2, 3\}$$

$$B = \{3, 4, 5\}$$

Thus,

$$A \cup B = \{1, 2, 3, 4, 5\}$$

The element 3 is written only once.

Empty Intersection

If two sets have no common elements, then their intersection is the **empty set**, written as $\emptyset$.

$$A = \{1, 2, 3\}$$

$$B = \{4, 5, 6\}$$

Therefore,

$$A \cap B = \emptyset$$

Venn Diagrams

Venn diagrams help visualize the union and intersection of sets.

The union includes all elements in both sets.

The intersection includes only elements common to both sets.

Examples

Example 1

Let

$$A = \{1, 2, 3, 4\}$$

and

$$B = \{3, 4, 5, 6\}.$$

Then

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

and

$$A \cap B = \{3, 4\}.$$

Example 2

Let

$$M = \{a, e, i, o\}$$

and

$$N = \{i, o, u\}.$$

Then

$$M \cup N = \{a, e, i, o, u\}$$

and

$$M \cap N = \{i, o\}.$$

Example 3

Let

$$P = \{2, 4, 6, 8\}$$

and

$$Q = \{1, 3, 5, 7\}.$$

Then

$$P \cup Q = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

and

$$P \cap Q = \emptyset.$$

Practice Exercises

Practice 1

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$.

Find $A \cup B$ and $A \cap B$.

Practice 2

Let $M = \{a, e, i, o\}$ and $N = \{i, o, u\}$.

Find $M \cup N$ and $M \cap N$.

Practice 3

Let $P = \{2, 4, 6, 8\}$ and $Q = \{1, 3, 5, 7\}$.

Find $P \cup Q$ and $P \cap Q$.

Practice 4

Let $X = \{5, 10, 15, 20\}$ and $Y = \{10, 20, 30\}$.

Find $X \cup Y$ and $X \cap Y$.

Practice 5

Let $A = \{m, n, p, q\}$ and $B = \{p, q, r, s\}$.

Find $A \cup B$ and $A \cap B$.

Practice 6

Let $C = \{7, 8, 9, 10\}$ and $D = \{8, 10, 12\}$.

Find $C \cup D$ and $C \cap D$.

Practice 7

Let $R = \{2, 3, 5, 7, 11\}$ and $S = \{5, 7, 13\}$.

Find $R \cup S$ and $R \cap S$.

Practice 8

Let $E = \{x, y, z\}$ and $F = \{y, z, w\}$.

Find $E \cup F$ and $E \cap F$.

Practice 9

Let $G = \{12, 14, 16\}$ and $H = \{14, 18, 20\}$.

Find $G \cup H$ and $G \cap H$.

Practice 10

In a class survey,

$$A = \{\text{Ana, Ben, Carl, Dana}\}$$

like basketball and

$$B = \{\text{Carl, Dana, Ella, Fred}\}$$

like volleyball.

Find $A \cup B$ and $A \cap B$.

Activities

Activity 1

Let $A = \{2, 4, 6, 8\}$ and $B = \{6, 8, 10, 12\}$.

Find $A \cup B$ and $A \cap B$.

Activity 2

Let $M = \{a, b, c, d\}$ and $N = \{c, d, e, f\}$.

Find $M \cup N$ and $M \cap N$.

Activity 3

Let $P = \{9, 11, 13\}$ and $Q = \{2, 4, 6, 8\}$.
Find $P \cup Q$ and $P \cap Q$.

Activity 4

Let $X = \{15, 20, 25\}$ and $Y = \{20, 30, 35\}$.
Find $X \cup Y$ and $X \cap Y$.

Activity 5

Let $A = \{h, i, j, k\}$ and $B = \{j, k, l, m\}$.

Find $A \cup B$ and $A \cap B$.

Activity 6

Let $C = \{14, 16, 18, 20\}$ and $D = \{16, 18, 22\}$.

Find $C \cup D$ and $C \cap D$.

Activity 7

Let $R = \{1, 2, 3, 5, 8\}$ and $S = \{3, 5, 13\}$.

Find $R \cup S$ and $R \cap S$.

Activity 8

Let $E = \{r, s, t\}$ and $F = \{t, u, v\}$.

Find $E \cup F$ and $E \cap F$.

Activity 9

Let $G = \{21, 24, 27\}$ and $H = \{24, 27, 30\}$.
Find $G \cup H$ and $G \cap H$.

Activity 10

In a class survey,

$$A = \{\text{Grace, Henry, Ivan, Joy}\}$$

like reading and

$$B = \{\text{Ivan, Joy, Kate, Liam}\}$$

like drawing.

Find $A \cup B$ and $A \cap B$.

Thank you!